

Achieve asymptotic three-quarters approximation for impartial selection in single-nomination graphs

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Website #13 | Solver verdict: PARTIAL | Verifier verdict: n/a | Shown: solver output

Problem

For each $n \geq 2$, let $\mathcal{G}_n$ denote the set of loopless directed graphs on $V = [n]$ in which every vertex has outdegree exactly 1. For an impartial mechanism f on $\mathcal{G}_n$, define its worst-case guarantee by

$$\text{perf}(f) := \inf_{G \in \mathcal{G}_n, \Delta(G) > 0} \frac{\sum_{v \in V} f_v(G) \delta^-(v, G)}{\Delta(G)}.$$

Let

$$\alpha_n^* := \sup\{\text{perf}(f) : f \text{ is impartial on } \mathcal{G}_n\}.$$

The exact question is whether

$$\liminf_{n \rightarrow \infty} \alpha_n^* \geq \frac{3}{4}.$$

A negative answer would require a fixed $\varepsilon > 0$ and infinitely many n such that

$$\alpha_n^* \leq \frac{3}{4} - \varepsilon.$$

A positive answer would require an explicit sequence of impartial mechanisms f_n with

$$\text{perf}(f_n) \geq \frac{3}{4} - o(1).$$

Alternative readings considered

The statement is essentially unambiguous. A weaker reading would ask only for a fixed- n obstruction or for a uniform-in- n upper bound. I pursue the strongest asymptotic reading, but the theorem proved below is only a fixed- n obstruction of size $O(1/n)$.

Related literature found

- No source I found is classified **SOLVES**; among the sources I located, the exact asymptotic $3/4$ question remains unresolved.
- Holzman–Moulin (2013) — **BACKGROUND**. This is the original single-nomination paper; it formulates impartial nomination/selection in the outdegree-1 setting and proves strong deterministic impossibility phenomena.
- Fischer–Klimm (2015) — **PARTIAL**. It established the earlier interval $[67/108, 35/48]$ for the single-nomination model and introduced/generalized the permutation viewpoint used throughout this area.
- Bousquet–Norin–Vetta (2014) — **PARTIAL**. It proves near- $3/4$ (and even near-1) guarantees when the maximum indegree Δ is sufficiently large, but this does not settle the worst-case-over-all-graphs problem because small Δ remains the bottleneck.

- Cembrano–Fischer–Klimm (2025) — **PARTIAL**. This is the closest published result I found: it gives the best published all- n lower bound 2105/3147, proves the same finite- n upper bound that I rederive below, and explicitly notes that the upper bounds still tend to $3/4$, leaving the asymptotic $3/4$ question open.
- Cembrano–Fischer–Klimm (2025), *Optimal Impartial Correspondences* — **BACKGROUND**. This looked promising because it studies impartiality under bounded outdegree, but it concerns subset-valued correspondences and a different objective (minimum indegree among selected vertices), so it does not directly answer the one-winner multiplicative problem here.

Proof sketch

I prove a sharp finite- n upper bound:

$$\alpha_n^* \leq \frac{3n^3 - 19n^2 + 30n - 4}{4n(n-2)(n-4)} = \frac{3}{4} - \Theta\left(\frac{1}{n}\right).$$

This does *not* settle the asymptotic question, but it is a strong general obstruction.

The proof is a linear-constraint argument built from a family of graphs $G_0, G_1, \dots, G_{n'}$ with a 2-cycle $1 \leftrightarrow 2$ and two directed paths feeding into the cycle. The split point between the two paths slides with i . Because the mechanism is impartial and (after symmetrization) symmetric, probabilities of certain vertices across this family must coincide. I encode these recurring probabilities by variables p_v and x_i .

The family is designed so that:

1. normalization on the graphs G_i yields a system of linear inequalities involving the p_v 's and x_i 's;
2. summing these inequalities forces at least one x_i to be at most $1/2 - O(1/n)$;
3. a single edge redirection turns G_i into a graph G'_i with a *unique* maximum-indegree vertex of indegree 2, whose selection probability is exactly x_i by impartiality.

On such a graph G'_i , the expected selected indegree is at most $2x_i + 1(1 - x_i) = 1 + x_i$, so the approximation ratio is at most $(1 + x_i)/2$. Plugging in the bound on $\min_i x_i$ gives the theorem.

Main result

Theorem 1. *For every $n \geq 6$,*

$$\alpha_n^* \leq \frac{3n^3 - 19n^2 + 30n - 4}{4n(n-2)(n-4)} = \frac{3}{4} - \frac{n^2 - 6n + 4}{4n(n-2)(n-4)} < \frac{3}{4}.$$

In particular,

$$\alpha_7^* \leq \frac{76}{105}.$$

Proof

Lemma 1 (Symmetrization). *Let f be an impartial mechanism on $\mathcal{G}_n$. Define f^s by*

$$f_v^s(G) := \frac{1}{n!} \sum_{\pi \in S_n} f_{\pi(v)}(G^\pi),$$

where G^π is obtained from G by relabeling every vertex u as $\pi(u)$. Then:

1. f^s is symmetric;
2. if f is impartial, then f^s is impartial;
3. if f is α -optimal on $\mathcal{G}_n$, then f^s is α -optimal on $\mathcal{G}_n$.

Proof. For symmetry, let $\sigma \in S_n$. Then

$$f_{\sigma(v)}^s(G^\sigma) = \frac{1}{n!} \sum_{\pi \in S_n} f_{\pi(\sigma(v))}((G^\sigma)^\pi) = \frac{1}{n!} \sum_{\pi \in S_n} f_{(\pi \circ \sigma)(v)}(G^{\pi \circ \sigma}).$$

As π ranges over S_n , so does $\pi \circ \sigma$. Therefore

$$f_{\sigma(v)}^s(G^\sigma) = f_v^s(G).$$

For impartiality, suppose $G, G' \in \mathcal{G}_n$ differ only in the outgoing edge of vertex v . Then for every $\pi \in S_n$, the relabeled graphs $G^\pi, (G')^\pi$ differ only in the outgoing edge of $\pi(v)$. Since f is impartial,

$$f_{\pi(v)}(G^\pi) = f_{\pi(v)}((G')^\pi).$$

Averaging over π gives $f_v^s(G) = f_v^s(G')$.

For α -optimality, fix $G \in \mathcal{G}_n$. Because relabeling preserves indegrees and the value of Δ ,

$$\sum_{v=1}^n f_v^s(G) \delta^-(v, G) = \frac{1}{n!} \sum_{\pi \in S_n} \sum_{v=1}^n f_{\pi(v)}(G^\pi) \delta^-(v, G) = \frac{1}{n!} \sum_{\pi \in S_n} \sum_{u=1}^n f_u(G^\pi) \delta^-(u, G^\pi).$$

If f is α -optimal, each inner sum is at least $\alpha \Delta(G^\pi) = \alpha \Delta(G)$. Hence

$$\sum_{v=1}^n f_v^s(G) \delta^-(v, G) \geq \alpha \Delta(G).$$

So f^s is α -optimal. □

Proof of the theorem. Fix $n \geq 6$, and let f be an impartial α -optimal mechanism on $\mathcal{G}_n$. By the symmetrization lemma, we may assume f is symmetric.

Let

$$n' := \left\lfloor \frac{n}{2} \right\rfloor - 1.$$

For a graph H and a vertex v , write H^{-v} for the digraph obtained from H by deleting the unique outgoing edge of v .

Define the following graphs on $[n]$.

- G_0 : edges

$$(1, 2), (2, 1), (3, 2), (4, 3), \dots, (n, n-1).$$

- For each $i \in \{1, \dots, n'\}$, G_i : edges

$$(1, 2), (2, 1), (3, 1), (i+3, 2),$$

together with

$$(v, v-1) \quad (4 \leq v \leq n, v \neq i+3).$$

Thus G_i consists of the 2-cycle $1 \leftrightarrow 2$, a left path

$$i+2 \rightarrow i+1 \rightarrow \dots \rightarrow 4 \rightarrow 3 \rightarrow 1,$$

and a right path

$$n \rightarrow n-1 \rightarrow \dots \rightarrow i+4 \rightarrow i+3 \rightarrow 2.$$

- C_n : the directed n -cycle

$$1 \leftarrow 2 \leftarrow 3 \leftarrow \dots \leftarrow n \leftarrow 1.$$

- $C_{2,n}$: a 2-cycle on $\{1, 2\}$ and a directed cycle on $\{3, \dots, n\}$, i.e.

$$(1, 2), (2, 1), (3, n), (4, 3), \dots, (n, n-1).$$

Set

$$p_v := f_v(G_0) \quad (1 \leq v \leq n),$$

and for $1 \leq i \leq n'$,

$$x_i := f_2(G_i).$$

We first record one reusable observation.

Observation. If $G, H \in \mathcal{G}_n$, $v, w \in [n]$, and $\pi \in S_n$ satisfy $\pi(v) = w$ and

$$(G^{-v})^\pi = H^{-w},$$

then $f_v(G) = f_w(H)$.

Indeed, under the displayed hypothesis, the full graphs G^π and H differ only in the outgoing edge of w . By impartiality,

$$f_w(G^\pi) = f_w(H).$$

By symmetry of f ,

$$f_v(G) = f_w(G^\pi).$$

Hence $f_v(G) = f_w(H)$, as claimed.

We now derive the identities and inequalities needed for the upper bound.

Step 1: two base probabilities. Because $G_0^{-1} = C_n^{-1}$, the observation gives

$$p_1 = f_1(C_n).$$

In C_n , every vertex lies in the same orbit under a cyclic relabeling preserving the graph. Since f is symmetric, all vertices of C_n have the same selection probability, and these probabilities sum to 1. Therefore

$$p_1 = \frac{1}{n}. \quad (1)$$

Similarly, $G_0^{-3} = C_{2,n}^{-3}$, so

$$p_3 = f_3(C_{2,n}).$$

In $C_{2,n}$, the permutation fixing 1, 2 and cycling $3, 4, \dots, n$ preserves the graph, so

$$f_3(C_{2,n}) = f_4(C_{2,n}) = \dots = f_n(C_{2,n}).$$

These $n - 2$ equal probabilities sum to at most 1, hence

$$p_3 \leq \frac{1}{n-2}. \quad (2)$$

Step 2: linking $f_2(G_i)$ to $f_1(G_{i+1})$. For $0 \leq i \leq n' - 1$, define $\sigma_i \in S_n$ by

$$\sigma_i(v) = \begin{cases} 3, & v = 1, \\ 1, & v = 2, \\ v + 1, & 3 \leq v \leq i + 2, \\ 2, & v = i + 3, \\ v, & v \geq i + 4. \end{cases}$$

This permutation matches the two inward-directed paths of G_i^{-2} to those of G_{i+1}^{-1} , so

$$(G_i^{-2})^{\sigma_i} = G_{i+1}^{-1}.$$

By the observation,

$$f_2(G_i) = f_1(G_{i+1}) \quad (0 \leq i \leq n' - 1). \quad (3)$$

In particular,

$$f_1(G_1) = f_2(G_0) = p_2,$$

and for $1 \leq i \leq n' - 1$,

$$f_1(G_{i+1}) = x_i.$$

Step 3: identifying two more probabilities in each G_i . Fix $i \in \{1, \dots, n'\}$.

First define $\tau_i \in S_n$ by

$$\tau_i(v) = \begin{cases} v, & v \in \{1, 2\}, \\ v + n - i - 2, & 3 \leq v \leq i + 2, \\ v - i, & i + 3 \leq v \leq n. \end{cases}$$

This shifts the left branch of G_i to the far end of the long path in G_0 , and one checks edge-by-edge that

$$(G_i^{-3})^{\tau_i} = G_0^{-(n-i+1)}.$$

Hence

$$f_3(G_i) = p_{n-i+1}. \quad (4)$$

Second, let $\rho = (1\ 2)$ be the transposition swapping 1 and 2. After deleting the outgoing edge of $i + 3$, the only difference between G_i and G_0 is whether the left branch hits 1 or 2, so

$$(G_i^{-(i+3)})^\rho = G_0^{-(i+3)}.$$

Therefore

$$f_{i+3}(G_i) = p_{i+3}. \quad (5)$$

Step 4: normalization inequalities. From $\sum_{v=1}^n p_v = 1$, together with (1) and (2), we obtain

$$p_2 + \sum_{v=4}^n p_v = 1 - p_1 - p_3 \geq 1 - \frac{1}{n} - \frac{1}{n-2} = \frac{n^2 - 4n + 2}{n(n-2)}. \quad (6)$$

For brevity set

$$A := \frac{n^2 - 4n + 2}{n(n-2)}.$$

Now consider G_1 . The probability of vertex 1 is p_2 by (3) with $i = 0$; the probability of vertex 2 is x_1 by definition; the probabilities of vertices 3 and 4 are p_n and p_4 by (4) and (5). Since total probability is 1,

$$p_2 + p_4 + p_n + x_1 \leq 1. \quad (7)$$

Likewise, for each $i \in \{2, \dots, n'\}$, in the graph G_i the probabilities of vertices $1, 2, 3, i + 3$ are $x_{i-1}, x_i, p_{n-i+1}, p_{i+3}$, respectively. Hence

$$p_{n-i+1} + p_{i+3} + x_{i-1} + x_i \leq 1 \quad (2 \leq i \leq n'). \quad (8)$$

Step 5: summing the inequalities.

Case 1: n odd. Then $n' = (n - 3)/2$. The indices

$$i + 3 \quad (2 \leq i \leq n')$$

run through

$$5, 6, \dots, \frac{n+3}{2},$$

while the indices

$$n - i + 1 \quad (2 \leq i \leq n')$$

run through

$$\frac{n+5}{2}, \frac{n+7}{2}, \dots, n-1.$$

Together with the terms p_2, p_4, p_n from (7), every variable in

$$p_2, p_4, p_5, \dots, p_n$$

appears exactly once in the sum of (7) and all instances of (8). Therefore

$$p_2 + \sum_{v=4}^n p_v + 2 \sum_{i=1}^{n'-1} x_i + x_{n'} \leq n'. \quad (9)$$

Combining (9) with (6) gives

$$2 \sum_{i=1}^{n'-1} x_i + x_{n'} \leq n' - A = \frac{n-3}{2} - A. \quad (10)$$

Case 2: n even. Then $n' = n/2 - 1$. Now the two index families in (8) overlap exactly once:

$$n - n' + 1 = n' + 3 = \frac{n}{2} + 2.$$

To account for this single repeated p -variable, take $2 \times (7)$, add $2 \times (8)$ for $i = 2, \dots, n' - 1$, and add $1 \times (8)$ for $i = n'$. On the p -side this yields

$$2p_2 + 2 \sum_{v=4}^n p_v,$$

and on the x -side it yields

$$4 \sum_{i=1}^{n'-2} x_i + 3x_{n'-1} + x_{n'}.$$

Hence

$$2p_2 + 2 \sum_{v=4}^n p_v + 4 \sum_{i=1}^{n'-2} x_i + 3x_{n'-1} + x_{n'} \leq 2n' - 1. \quad (11)$$

Using $2 \times (6)$, we obtain

$$4 \sum_{i=1}^{n'-2} x_i + 3x_{n'-1} + x_{n'} \leq 2n' - 1 - 2A = n - 3 - 2A. \quad (12)$$

Step 6: turning one x_i into an approximation-ratio upper bound. For $i \in \{1, \dots, n'\}$, let G'_i be obtained from G_i by changing the outgoing edge of vertex 2 from 1 to n .

By impartiality, the selection probability of vertex 2 does not change:

$$f_2(G'_i) = f_2(G_i) = x_i. \quad (13)$$

In G'_i , vertex 2 has indegree 2 (from 1 and $i + 3$); every other vertex has indegree at most 1. Thus

$$\Delta(G'_i) = 2, \quad \arg \max_v \delta^-(v, G'_i) = \{2\}.$$

Therefore the expected indegree selected by f on G'_i is at most

$$2x_i + 1 \cdot (1 - x_i) = 1 + x_i.$$

Since f is α -optimal,

$$\alpha \leq \frac{1 + x_i}{2} \quad (1 \leq i \leq n'). \quad (14)$$

Hence

$$\alpha \leq \frac{1 + \min_{1 \leq i \leq n'} x_i}{2}. \quad (15)$$

Step 7: bounding $\min_i x_i$.

If n is odd, then from (10),

$$\min_{1 \leq i \leq n'} x_i \leq \frac{2 \sum_{i=1}^{n'-1} x_i + x_{n'}}{2n' - 1} \leq \frac{\frac{n-3}{2} - A}{n-4} = \frac{n-3-2A}{2(n-4)}. \quad (16)$$

If n is even, then from (12),

$$\min_{1 \leq i \leq n'} x_i \leq \frac{4 \sum_{i=1}^{n'-2} x_i + 3x_{n'-1} + x_{n'}}{4n' - 4} \leq \frac{n-3-2A}{2(n-4)}. \quad (17)$$

Thus in both parity cases,

$$\min_{1 \leq i \leq n'} x_i \leq \frac{n-3-2A}{2(n-4)}. \quad (18)$$

Plugging (18) into (15),

$$\alpha \leq \frac{1}{2} \left(1 + \frac{n-3-2A}{2(n-4)} \right).$$

Substituting $A = \frac{n^2 - 4n + 2}{n(n-2)}$ and simplifying,

$$\alpha \leq \frac{3n^3 - 19n^2 + 30n - 4}{4n(n-2)(n-4)}.$$

Since f was an arbitrary impartial α -optimal mechanism, this proves

$$\alpha_n^* \leq \frac{3n^3 - 19n^2 + 30n - 4}{4n(n-2)(n-4)}.$$

Finally,

$$\frac{3}{4} - \frac{3n^3 - 19n^2 + 30n - 4}{4n(n-2)(n-4)} = \frac{n^2 - 6n + 4}{4n(n-2)(n-4)} > 0$$

for every $n \geq 6$, so the bound is strictly smaller than $3/4$. For $n = 7$ it becomes

$$\frac{3 \cdot 7^3 - 19 \cdot 7^2 + 30 \cdot 7 - 4}{4 \cdot 7 \cdot 5 \cdot 3} = \frac{76}{105}.$$

□

Gap to the full statement

The theorem proves only

$$\alpha_n^* \leq \frac{3}{4} - \Theta\left(\frac{1}{n}\right),$$

so it does *not* exclude the possibility that $\alpha_n^* \rightarrow 3/4$ as $n \rightarrow \infty$. To settle the original problem one still needs one of the following:

- a construction of impartial mechanisms f_n with $\text{perf}(f_n) \geq 3/4 - o(1)$; or
- a fixed $\varepsilon > 0$ and infinitely many n with $\alpha_n^* \leq 3/4 - \varepsilon$.

What the proof *does* show is that any eventual $3/4 - o(1)$ positive construction must already be nearly optimal on a very rigid family of $\Delta = 2$ witness graphs built from a 2-cycle with two incoming paths.

Among the sources I found, the best published all- n lower bound is 2105/3147, while the best published general upper bounds still approach $3/4$.

Self red-team

- **Counterexample attempts to the symmetrization lemma.** I tried to break impartiality under averaging over relabelings. The only possible issue would be that relabeling changes which vertex is “the same” agent; the proof explicitly tracks this via $\pi(v)$, so the argument is sound.
- **Counterexample attempts to the deleted-edge observation.** A natural objection is that f is not defined on graphs with a deleted outgoing edge. The proof does *not* evaluate f there: the deleted-edge graphs are only witnesses that two full graphs differ in one agent’s outgoing edge, which is exactly the domain where impartiality applies.
- **Boundary checks.** For $n = 6$, the formula gives $35/48$. For $n = 7$, it gives $76/105$. I checked both odd/even summation patterns separately. The even case $n = 6$ is the smallest one and is the most delicate because $n' = 2$; the weighted sum in (11)–(12) still works and reduces to the advertised coefficients $3, 1$ on x_1, x_2 .
- **Index-counting objection.** The most plausible bookkeeping error is miscounting the p_v -variables when summing (7) and (8). I explicitly checked:

$$\{i + 3 : 2 \leq i \leq n'\} = \{5, \dots, \lfloor n/2 \rfloor + 2\},$$

$$\{n - i + 1 : 2 \leq i \leq n'\} = \{\lceil n/2 \rceil + 2, \dots, n - 1\},$$

with exactly one overlap when n is even, namely at $n/2 + 2$. That is exactly why the even case needs asymmetric weights.

- **Does the proof contradict known near-3/4 high- Δ results?** No. The obstruction here uses only $\Delta = 2$ graphs. Results that become strong when Δ is large are therefore perfectly compatible with this proof.
- **Does the proof accidentally settle the asymptotic question negatively?** No. The explicit gap from $3/4$ is

$$\frac{n^2 - 6n + 4}{4n(n - 2)(n - 4)} = \Theta(1/n),$$

which vanishes as $n \rightarrow \infty$.

Ideas tried

1. **Reweight the same two-arm family to force a constant gap below $3/4$.**

Idea: optimize the linear-combination coefficients in the G_i -inequalities more aggressively, hoping to show $\min_i x_i \leq 1/2 - \varepsilon$ for a fixed $\varepsilon > 0$.

Obstruction: the normalization constraints only control roughly one unit of p -mass spread across $\Theta(n)$ graphs, and every weighting I could justify still led to $\min_i x_i \leq 1/2 - O(1/n)$, not a constant gap.

Next step: find a witness family where each normalization inequality contains three or more x -variables, not just two consecutive ones.

2. **Replace the 2-cycle family by a 3-cycle / three-arm family.**

Idea: after modification, produce a graph with a unique indegree-3 optimum and derive an

upper bound of the form $\alpha \leq (2 + \min y_i)/3$; to beat $3/4$, one would need $\min y_i < 1/4$.

Obstruction: I could not force such a small y_i using hand-derived constraints. The analogous counting seemed too weak, and I did not obtain a clean explicit family with tractable equalities like (3)–(5).

Next step: formulate the three-arm version as a reduced LP on orbit representatives and search for a dual certificate.

3. Positive-side construction attempts via simple mixtures.

Idea: mix permutation-like rules with very local impartial rules such as “pick the nominee of a random vertex” or “pick a cycle vertex according to a local score” to improve behavior on the degree-2 obstruction family.

Obstruction: simple local rules either give poor worst-case performance on long-path witnesses or fail to preserve impartiality after tie-breaking adjustments. Convex combinations of such rules did not visibly move the bottleneck above the known $> 2/3$ regime.

Next step: a successful $3/4$ -construction likely needs a genuinely nonlocal mechanism that recognizes and handles the two-cycle/path witnesses.

4. Yao-style distributional upper bound.

Idea: encode impartiality as linear constraints on $p_v(G)$, put a distribution on functional digraphs, and search for a dual witness proving a strict asymptotic gap below $3/4$.

Obstruction: without a strong structural ansatz, the state space of deleted-edge neighborhoods becomes too large to control cleanly by hand.

Next step: quotient the LP by relabeling orbits of functional digraphs and search for a small extremal family.

Invoked results

No external theorem is used as a step inside the proof. The proof is self-contained. The web was used only for orientation, for checking what is already known, and for verifying bibliographic data.

Self-confidence

3 — I checked each combinatorial step and the algebra carefully, and I could not break the proof, but long index bookkeeping always leaves some residual risk of a small slip.

Bibliography

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\begin{verbatim}
@article{HolzmanMoulin2013,
  author = {Holzman, Ron and Moulin, Herv{'e}},
  title = {Impartial Nominations for a Prize},
  journal = {Econometrica},
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  number = {1},
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  year = {2013},
  doi = {10.3982/ECTA10523}
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@article{FischerKlimm2015,
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  title = {Optimal Impartial Selection},
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  number = {5},
  pages = {1263--1285},
  year = {2015},
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}

@misc{BousquetNorinVetta2014,
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  title = {A Near-Optimal Mechanism for Impartial Selection},
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@article{CembranoFischerKlimm2025,
  author = {Cembrano, Javier and Fischer, Felix and Klimm, Max},
  title = {Improved Bounds for Single-Nomination Impartial Selection},
  journal = {Mathematics of Operations Research},
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  number = {2},
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  year = {2025},
  doi = {10.1287/moor.2024.0431}
}

@article{CembranoFischerKlimm2025Correspondences,
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  title = {Optimal Impartial Correspondences},
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